

BAKAR QURESHI

MACHINE LEARNING ENGINEER

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SUMMARY

Software Engineering student at IIUI with hands-on experience across full-stack development and applied AI/ML. Completed Andrew Ng's Machine Learning Specialization — covering supervised & unsupervised learning, neural networks, and ML system design — and implemented a Vision Transformer (ViT) in a real-world project, exploring attention-based architectures for image understanding beyond traditional CNNs.

SKILLS

AI/ML: TensorFlow, Keras, PyTorch, Scikit-learn, NumPy, Pandas, Matplotlib, LangGraph, LangChain

Languages: C++, JavaScript (ES6+), Python

Frontend: React.js, Bootstrap, HTML, CSS

Backend: Node.js, Express.js, Flask

Tools & Environments: Git & GitHub, Jupyter Notebook, VS Code, WSL, Ubuntu (Linux), Hugging

Databases: MongoDB, MySQL

Concepts: DSA, OOP, REST APIs, Authentication, Responsive UI Design, Supervised & Unsupervised Learning, Neural Networks

PROJECTS

Solar Sentinel — PyTorch, ResNet, TorchVision, Transfer Learning, CNN

- Developed an image classification system to detect faults in solar panels across 6 fault categories.
- Leveraged a pretrained ResNet model with transfer learning and fine-tuning to overcome a limited dataset of 861 images.
- Applied data augmentation techniques to improve generalization, achieving 96.3%+ classification accuracy.

Drafter — LangGraph, LangChain, Groq, LLaMA 3.3 70B

- Built an agentic document writing assistant with **human-in-the-loop** review using LangGraph's interrupt mechanism.
- Implemented **intent detection** to infer user satisfaction from natural language — automatically saving and exiting when the user signals completion.

AI Code Reviewer & Debugger — LangChain, Python, Structured Output Parsing

- Built a CLI-based autonomous code review and debugging agent using LangChain and LLM-powered analysis.
- Implemented **automatic bug detection and in-place file modification** — agent identifies issues and applies fixes directly to source files.
- Used structured output parsing to deliver consistent, categorized feedback on code quality, bugs, and improvements

EDUCATION

Bachelor of Science in Software Engineering

International Islamic University Islamabad

2023-Present
